

Functional Equations

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1 Standard Tactics

- **Try to guess the functions!** This helps you know what to prove. **Always write down that they work**, as showing a solution is valid is often worth a point!
- **Plug in values** to make things cancel out. Always explicitly write down which **substitutions** you are making (e.g., "Let $x = 1$ and $y = f(z)$ ").
- **Look for symmetry**, for example, the fff trick: if $f(f(x)) = g(x)$, then $f(f(f(x))) = f(g(x)) = g(f(x))$.
- **Cauchy's Equation:** For $f : R \rightarrow R$ satisfying $f(x + y) = f(x) + f(y)$, it follows that $f(qx) = qf(x)$ for all $q \in Q$. Furthermore, if the function is **continuous, bounded, or monotonic**, then f is linear ($f(x) = cx$).
- **Defining a new function g** (e.g., $g(x) = f(x) - c$) to simplify the expression, then substituting back to solve for f .
- **Adding a free variable:** If $f(x + y) = g(x, y)$, then using the associative property $f(x + y + z) = g(x + z, y) = g(x, y + z)$ might be useful.
- **"Ectivities":** Check for **injectivity, surjectivity, and bijectivity**, as these properties often allow you to "cancel" the function from both sides of an equation.

To get better at this, do A LOT of FEs.

Flavour:

- **Continuity:** Monotonicity combined with surjectivity yields that f is continuous. You can then investigate limits such as $x \rightarrow 0^+$.
- **Fixed Points and Level Sets:** Some problems will require you to find sets of the form $\{x \mid f(x) = c\}$ for some constant c .
- **Periodicity:** Investigating periodicity is often useful when trying to prove or disprove injectivity.
- **Extrema:** You might find yourself using maxima/minima arguments to bound the function.

To get better, do A LOT of HARD FEs.

2 Group Solving

- **IMO17/2** Let R be the set of real numbers. Determine all functions $f : R \rightarrow R$ such that, for any real numbers x and y ,

$$f(f(x)f(y)) + f(x + y) = f(xy).$$

A hint will be given every 15 minutes.

3 Problems

For the last three, you can look at Hints then Solutions after a serious attempt. They're hard. The first two are much easier.

- **Socrates** Find all solutions $f : R^+ \rightarrow R^+$ satisfying the equation:

$$f(x+y)^2 = f(x)^2 + 2f(xy) + f(y)^2, \quad \forall x, y \in R^+$$

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- **Continuity** Find all functions $f : (0, +\infty) \rightarrow (0, +\infty)$, such that, for all $x, y > 0$, the following equation holds:

$$f(xy + f(x)) = xf\left(\frac{1}{2} + f\left(y + \frac{1}{2}\right)\right)$$

.

- **APMO16** Find all functions $f : R^+ \rightarrow R^+$ such that

$$(z+1)f(x+y) = f(xf(z)+y) + f(yf(z)+x),$$

for all positive real numbers x, y, z .

- **Iran19** Find all functions $f : R \rightarrow R$ such that for all $x, y \in R$:

$$f\left(f(x)^2 - y^2\right)^2 + f(2xy)^2 = f\left(x^2 + y^2\right)^2$$

- **ISL07** Find all functions $f : R^+ \rightarrow R^+$ satisfying $f(x + f(y)) = f(x + y) + f(y)$ for all pairs of positive reals x and y . Here, R^+ denotes the set of all positive reals.

4 Hints

- Socrates: fff trick.
- Continuity: prove that f is continuous, use some very small ϵ to kill the problem.
- APMO16: Solve $f(x)=1$, Prove surjectivity. Finish off with a maximality argument.
- Iran19: you should have a formula that, if f^2 is injective at zero would finish the problem. Investigate the set $\{x|g(x) = 0\}$ to prove injectivity at zero, try to use what you know about AMGM to make a bound on the size of this set.
- ISL07: Show it's additive (Cauchy) and use the fact that it's bounded to prove it's linear. You can also prove it's injective through a periodicity argument.

Functional Equations Beyond Assertions

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Solutions usually go the following way:

- Make things cancel, exploit manifestations of symmetry...
- Try to find some values for specific constants $f(0)$ for $\mathbb{R}$ for example or $f(1)$ for $\mathbb{R}^+$ etc...
- The set $\{x|f(x) = c\}$ for some c .
- Induct on $\mathbb{Q}$ maybe...

When stuck, and you feel like assertions are no longer working it might be helpful to:

- Try and investigate (local or universal) injectivity using $f(a) = f(b), a \leq b$
- idem for surjectivity.
- Prove that if a property works for $z \Rightarrow$ it works for $cz \forall c$ as well; or $z_1 + z_2$ if it works for both z_1 and z_2 .

We can keep going forever but there's no magic recipe, you just need to do more (hard) problems.

1 Problems

1. **Kazakhstan Regional 26.** Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that for all real numbers x, y the equality

$$f(f(y) + x - y) + f(x - y) = f(xf(y) - y)$$

holds.

2. **Japan EGMO Preliminary 24.** Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(f(x) + xy) = f(x)f(x + y)$$

for all real numbers x, y .

3. **KosovoMO26.** Let $\mathbb{R}^+$ be the set of positive real numbers. Find all functions $f : \mathbb{R}^+ \rightarrow \mathbb{R}^+$ such that for all $x, y \in \mathbb{R}^+$,

$$f(f(x^2y) + xf(y)^2) = xyf(x + f(y)).$$

4. **ELMO23/A2.** Let $\mathbb{R}_{>0}$ denote the set of positive real numbers. Find all functions $f : \mathbb{R}_{>0} \rightarrow \mathbb{R}_{>0}$ such that for all positive real numbers x and y ,

$$f(xy + 1) = f(x)f\left(\frac{1}{x} + f\left(\frac{1}{y}\right)\right).$$

Note: Always try to guess the solution set and show that said set satisfies the FE! It's worth a point most of the time.

2 Solutions

1. $P(0, 0)$ then $P(x, z)$ such that $f(z) = 0$.
2. $P(0, y) \Rightarrow f(0) = 0$. $P(z, y)$ such that $f(z) = 0 \Rightarrow z = 0$. $P(x, -x) \Rightarrow f$ constant or $f(x) = x^2$
3. $P(x, 1) \Rightarrow f$ is injective. We then do $P(x, \frac{1}{x})$ and $P(\frac{1}{x}, x)$ to find f either constant or *id*. Here, when solving, there's a moment you realize that you can't progress further through assertions, and you try to understand the behaviour of the function, leading you to investigate the 2-equation system.
4. $P(2, \frac{1}{2}) \Rightarrow \exists z f(z) = 1$ then $P(z, \frac{1}{z})$ and $P(1, \frac{1}{z}) \Rightarrow f(1) = 1..$
 $P(x, \frac{1}{x})$ gives us a motivation to go investigate injectivity.

We then investigate injectivity: let $f(a) = f(b) \Rightarrow f(ax+1) = f(bx+1)$ by $P(x, \frac{1}{a})$ and $P(x, \frac{1}{b})$. Here, one is attracted by this particular conclusion, and tries to turn it into a "either $\frac{1}{x}$ or constant."

$P(x, a)$ and $P(x, b)$ expand our previous result to include $f(a) = f(b) \Rightarrow f(ax) = f(bx)$. By taking $x := \frac{b}{a} \Rightarrow f(\frac{b}{a}) = 1 \Rightarrow f(x+1) = f(\frac{b}{a}x+1) \Rightarrow f(\frac{cx+1}{x+1}) = 1$ which either makes f constant or injective.